EARTH AND PLANETARY SCIENCE LETTERS 17 (1972) 217–226. NORTH-HOLLAND PUBLISHING COMPANY

THE PALAEOMAGNETISM OF SOME PROTEROZOIC–CAMBRIAN SEDIMENTS FROM THE AMADEUS BASIN, CENTRAL AUSTRALIA

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Received 12 June 1972
Revised version received 2 October 1972

A palaeomagnetic study of Proterozoic-Cambrian and Lower-Middle Cambrian sediments from central Australia has revealed that a polar shift (and/or plate movement) occurred over the time interval covered by the rock units sampled. The polar shift, from 8.5°N 35°W (α_{95} = 25°) to 11°N 37°E (α_{95} = 8.5°) closely matches a similar shift previously described from northern Australia and would suggest that the two crustal blocks have maintained their relative geographic positions since at least Early Cambrian times. Polarity data suggest that the geomagnetic field underwent a reversal during this period.

1. Introduction

This note describes first results from a palaeomagnetic research program designed to investigate the Lower Palaeozoic geomagnetic field and the relation of the major crustal blocks in Australia. Recent interpretation [1, 2] of the geological history of southeast Australia, in terms of plate tectonics, have stressed the complex tectonic history the region suffered during the Lower Palaeozoic. Therefore, to successfully describe an apparent polar wander path for Australia for this period, it would seem that results may be better obtained from localities situated in the older crustal regions. Indeed, almost all Lower Palaeozoic investigations have been carried out with material sampled from the Northern Territory and South Australia, viz. Cambrian [3, 6] and Ordovician [7]. If agreement between the palaeomagnetic records for the Lower Palaeozoic from separate crustal regions within Australia can be demonstrated, then the relationship of Australia to the rest of Gondwanaland may be clarified.

Geological mapping of the region known as the Amadeus Basin, central Australia, has recently been completed by the Australian Bureau of Mineral Resources. Various reports have been prepared by the Bureau (those of special interest were numbers 61, 85, 86 and 113) and recently, an account of the geology of the entire Basin has been published [8].

2. Geological note

The Amadeus Basin has been described [8] as a 500 mile-long intracratonic depression with an area of approximately 60 000 sq. miles. It contains Proterozoic, Cambrian, Ordovician, Silurian?, Devonian to Carboniferous and minor Permian deposits. It is estimated that 30 000 ft. of sedimentary rocks have been preserved. For palaeomagnetic investigation, rock samples were collected from the Late Precambrian and Palaeozoic sequence which is exposed in Ellery Creek located at 133°E 23.8°S. The Creek has cut a north–south valley through the steeply south-dipping beds of the Palaeozoic sequence along the northern edge of the Basin. Results from the Arumbera Sandstone and the Hugh River Shale are described in this paper. Both rock units dip uniformly southwards at about 70°.

Exposed stratigraphic levels of the upper 600 ft. or so of the Arumbera Sandstone yielded samples. It consists of red-brown feldspathic sandstone with a medium- to fine-grained texture. Current-bedding and

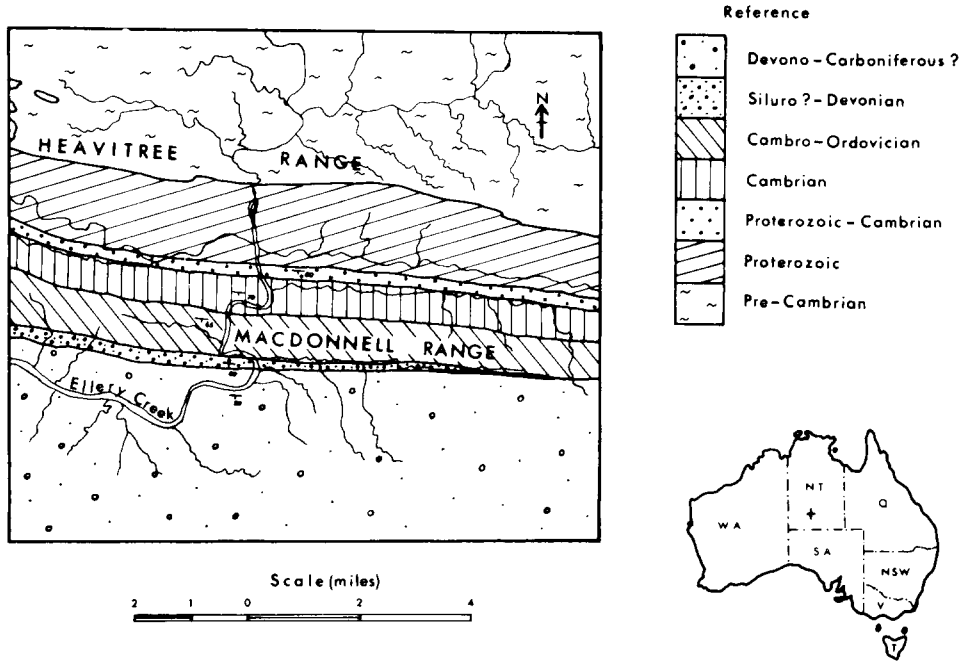


Fig. 1. A simplified geological map of the sampling locality. Samples were collested in the section following Ellery Creek. The locality is marked by a cross in the accompanying reference map of Australia.

slump-roll structures are commonly developed through the entire sequence.

The Hugh River Shale disconformably overlies the Arumbera Sandstone in Ellery Creek, but is conformably overlain by the Jay Creek Limestone, itself Middle Cambrian near the base. The formation consists of interbedded shale, siltstone, sandstone and carbonate beds, and since it is poorly exposed only 8 samples were obtained from 3 thin beds of silty mudstone near the base of the succession. No fossils, useful for geological dating, have been found in the Hugh River Shale; the assigned age [8] of Lower to Middle Cambrian is based on its stratigraphic position. A simplified geological map of the sampling locality and its position within Australia are shown in fig. 1.

3. Results from the Arumbera Sandstone

Specimens were cut from the 33 samples and each was measured (on a spinner magnetometer) for its direction and intensity of natural remanent magnetization (NRM) before cleaning was carried out. Within-sample scatter was often high (as can be inferred from a comparison of N and R in table 1) and the distribution of sample-means indicated no preferred sense of orientation, neither towards the direction of the present field nor towards a streaked distribution, so often apparent when rocks have undergone systematic partial remagnetization.

Stability of the magnetic remanence was tested using the partial thermal demagnetization technique [9] and zero magnetic field was maintained using the automatic field cancellation system built and described by McElhinny et al. [10]. Initial experimental procedure involved heating a pilot specimen from each of 10 samples (numbers A42 to A51 inclusive) to 200°C, and then in increments of 100°C to 600°C. Three further heatings were made between 600°C and 670°C. Stable magnetic remanence, if present, was usually revealed above 600°C, hence the necessity for higher temperature treatment to check the degree of repeatability of the results. Also on the basis of the initial experiments, steps 200°C and 300°C were omitted from subsequent thermal analyses since they revealed little or no information regarding the

Table 1

Arumbera Sandstone: Sample-mean data before and after partial thermal demagnetization. D and I are given with respect to bedding planes.

Sample	NRM Data					Cleaned Data			
	N	R	D	I	Int. $\times 10^{-6}$ Gauss	N	R	D	I
A19	3	2.810	349.5	+ 22.0	1.6	2	1.984	294.0	+ 36.0
A20	3	2.455	24.5	− 31.0	1.3	2	1.989	3.0	+ 43.0
A21	3	2.620	22.0	+ 31.5	0.7	2	1.936	65.0	+ 79.0
A22	3	2.315	345.0	+ 7.0	1.3	2	1.978	263.0	+ 52.0
A23	3	2.332	43.5	+ 51.5	3.9	2	1.718	32.0	+ 50.5
A24	2	1.779	334.0	+ 46.5	0.6				
A25	3	2.346	342.5	− 24.0	2.2				
A26	3	2.867	338.0	+ 2.5	3.1				
A27	3	2.192	340.5	+ 64.0	1.6	2	1.865	184.5	+ 57.5
A28	3	2.752	228.0	− 25.5	1.4	2	1.917	42.0	+ 75.0
A29	2	1.959	38.5	+ 58.0	5.7				
A30	3	2.617	9.0	+ 42.0	2.9	2	1.815	43.0	+ 54.5
A31	3	2.902	337.0	+ 30.0	2.2	2	1.938	240.0	+ 85.0
A32	3	2.519	294.0	+ 44.0	2.3	2	1.992	67.5	+ 67.0
A33	3	2.514	124.5	+ 56.5	4.6	2	1.972	130.5	+ 70.0
A34	3	2.202	225.0	− 41.0	3.6				
A35	3	2.109	295.0	− 31.0	2.6				
A36	3	1.450	285.0	− 21.0	1.6				
A37	2	1.793	272.0	− 65.0	1.7	2	1.872	175.0	+ 58.0
A38	3	1.868	315.5	− 25.5	1.7	2	1.870	3.5	+ 45.0
A39	3	2.798	121.0	+ 34.0	3.4	2	1.792	126.0	+ 76.0
A40	3	2.978	324.0	− 55.0	2.5				
A41	3	0.701	101.5	+ 28.5	1.4				
A42	3	2.929	140.5	− 69.0	9.6				
A43	3	2.959	156.0	− 65.0	9.0				
A44	3	2.995	340.5	− 15.0	1.4				
A45	3	2.937	342.0	− 13.5	1.8				
A46	2	1.957	337.0	− 17.0	14.3				
A47	3	2.979	332.0	− 35.0	15.1				
A48	3	2.999	37.5	− 6.0	36.2	2	1.795	33.0	+ 36.5
A49	3	2.999	20.5	− 11.0	26.8				
A50	3	2.618	110.0	− 26.5	6.1				
A51	3	1.705	286.5	+ 22.5	2.1	2	1.895	87.0	+ 78.0

Mean of sample-mean after cleaning

N	R	D	I	α_{95}
16	13.671	27.5	+ 78.5	15.8

Mean of VGP,s

N	R	lat.	long.	α_{95}
16	11.259	8.5° N	35° W	24.9

N: total number of specimens measured from each sample; R: resultant of N unit vectors; D: declination; I: inclination; Int: intensity of magnetization and α_{95} is the semi-angle of the cone of confidence at the 95% probability level (P = 0.05).
Directions of magnetization were analysed following the method of Fisher [15]. 'Cleaned' data lists only those results considered to be estimates of the primary direction of magnetization.

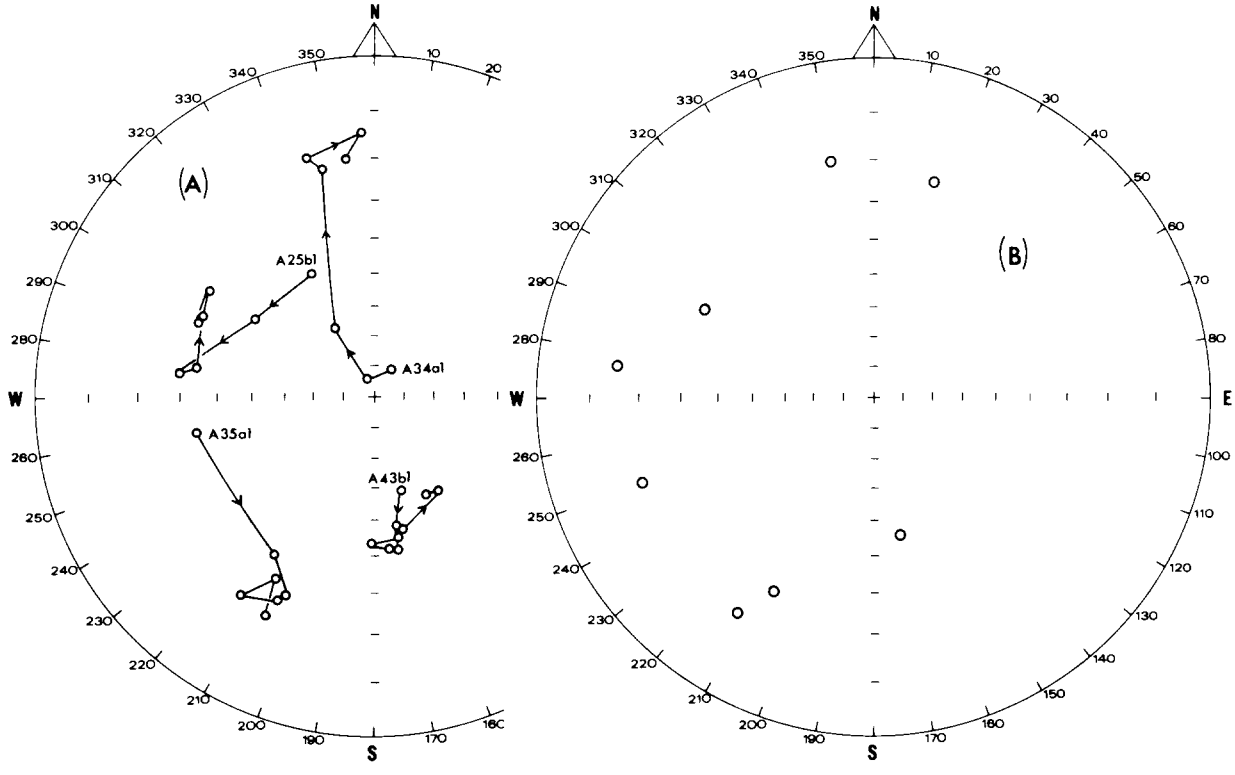


Fig. 2. Part A shows some examples of direction trends during treatment for specimens from the stable negative group. For the temperature at each step, refer to curves in fig. 4B. Although the high temperature directions appear to be stable, the sample-means (part B) remain strongly oblique to one another.

stable remanence. Thermal analysis of the NRM's of 23 remaining pilot specimens was completed following the modified procedure.

Nine specimens behaved in an unstable fashion. Their directions of magnetization, measured after each demagnetization step, were not repeatable and in the case of the samples A44 and A45, their magnetic moment was observed to alter during the actual measuring procedure. Specimens from these 9 samples were therefore not considered further. Response to thermal treatment of the 24 remaining specimens can be described in one of two ways. Firstly, 16 specimens changed their directions systematically and above approximately 600°C converged to a common axis with steep positive inclinations. The directions of magnetization of the other 8 specimens each indicated stability at high temperature, i.e., directions were repeatable; the stable directions, however, were scattered over one hemisphere of the stereogram (fig. 2B; results are plotted on a Wulff net) and displayed negative inclinations (pointing up). Examples of this behaviour are shown in fig. 2A. One further specimen was chosen from each of the 16 positively magnetized samples to serve as a check on the reliability of the pilot specimen data. They were heated and measured twice above 610°C and a sample-mean direction of magnetization was calculated from the specimen data. Individual specimen directions were obtained by taking the mean of the stable high-temperature directions. A summary of the data used in the analysis for palaeomagnetic pole computation is given in table 1; sample-means are also plotted in fig. 3.

Table 1 also lists the arithmetic-mean intensity for each sample; only 4 of the 33 samples had intensities an order of magnitude greater than 10^{-6} Gauss. Examples of the demagnetization-intensity curves, corresponding to the specimens plotted in fig. 2A are shown in fig. 4B; fig. 4A shows three examples of the

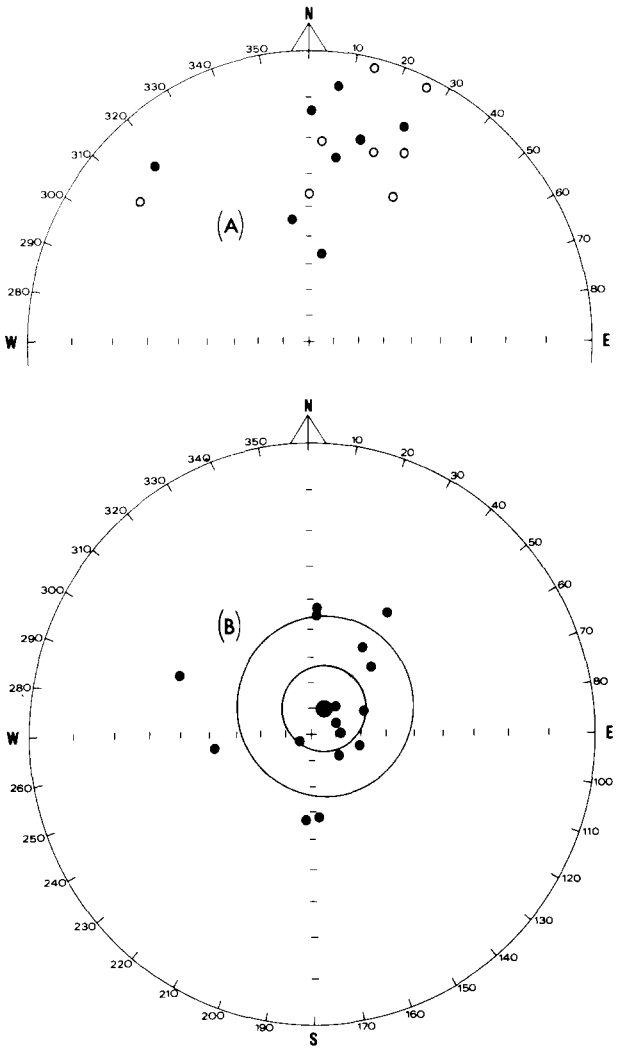


Fig. 3. High-temperature, sample-mean directions of magnetization in the Arumbera Sandstone: A, to present horizontal and B, with respect to bedding planes. The mean direction is represented by the large dot in B; the inner circle is the cone of confidence at the 95% probability level (α_{95}) and the outer circle is the circular standard deviation.

curves obtained from magnetically unstable specimens, and fig. 4C shows 4 curves relating to the stable positive samples. There is no common feature which could serve to distinguish one set of curves from any other. Thus, solely upon inspection, it was not possible to distinguish between the unstable, stable-positive and stable-negative responses to thermal treatment.

4. Results from the Hugh River Shale

Directions of magnetization obtained from each specimen before treatment are shown in fig. 5A and B. They form a streaked distribution along a great circle which passes through the present field axis, thus indicating that a proportion of their remanence was acquired in the direction of the present geomagnetic field. The fact that they streak to the east and west of the present field axis can be explained on the basis of the partial demagnetization results; both positive and negative polarities, referred to a common magnetization axis, are present. One sample, A56, was unstable; its magnetic moment was observed to vary during measurement. Specimens from all other samples produced stable directions above 500°C. The scatter of the positively and negatively magnetized groups decreased and they aligned approximately antiparallel. When referred to the present horizontal, fig. 5C, they are seen to lie remote from the present field axis and the streaking towards it has been eliminated. Arithmetic-mean intensities (NRM values) for all samples are listed in table 2.

5. Interpretation of results

(a) Arumbera Sandstone: The sample numbers, AI9–A51, give an indication of relative stratigraphic level, A19 was collected low in the succession and A51 was collected towards the top. However, the classification of a sample's behaviour into the unstable, stable-positive and stable-negative groups, does not seem to correlate with its stratigraphic position. Magnetically unstable samples were A24, 26, 29, 40, 41, 42, 44, 45 and A50; stable-negative samples were A25, 34, 35, 36, 43, 46, 47 and A49; stable-positive samples are listed in table 2 under cleaned data. It could be expected therefore, that any one or more of the stability categories may be encountered at arbitrary stratigraphic levels. Apparent instability of remanence may, in part, be due to the acquisition of spurious magnetic moments during the thermal experiments. If a weak stable component was present, it may have been masked by an acquired component, depending on the relative strengths of the moments involved. Low stability is a problem often encountered in palaeomagnetic studies whereas those samples classi-

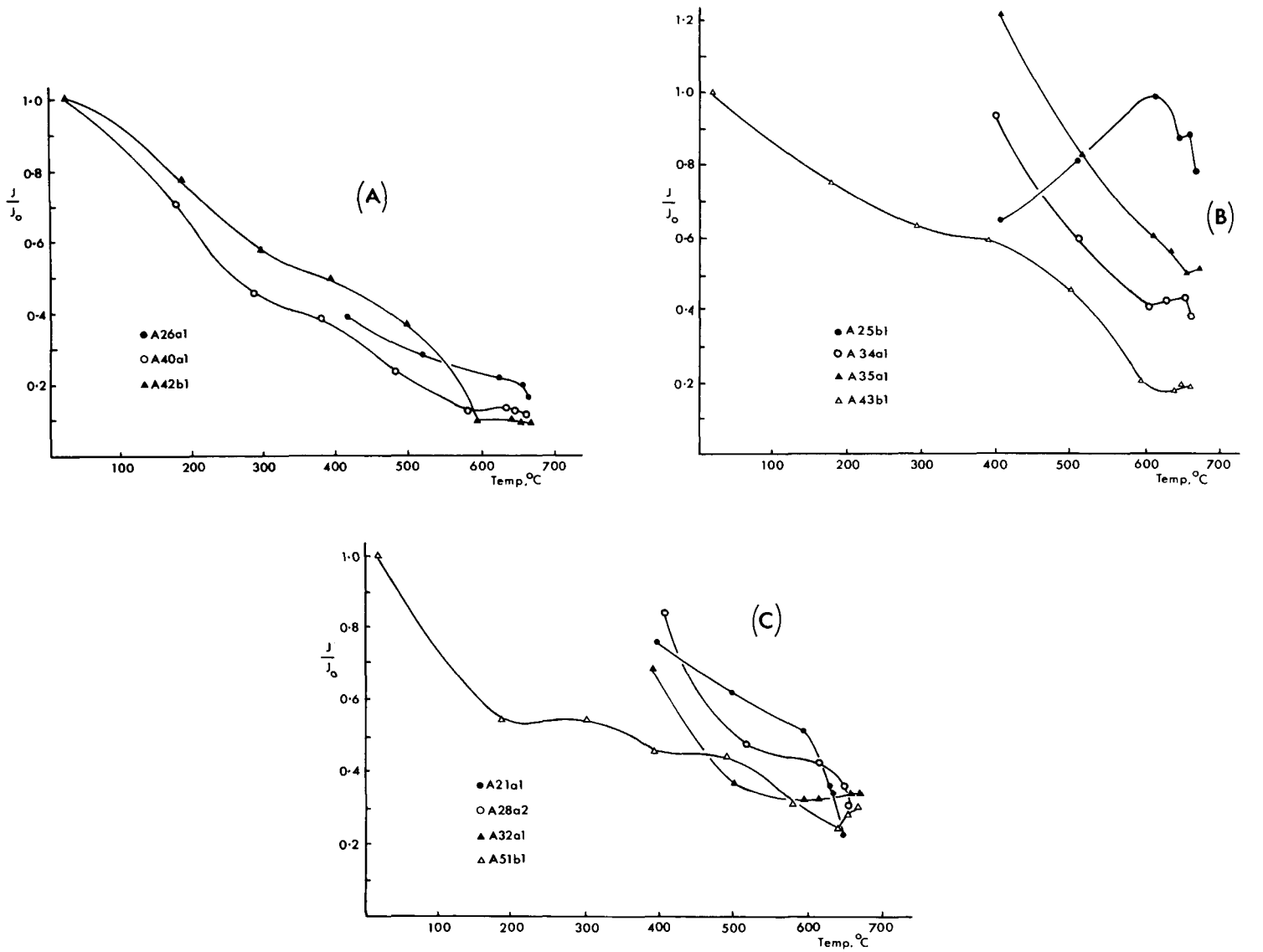


Fig. 4. Demagnetization-intensity curves for pilot specimens of Arumbera Sandstone. Part A shows three curves obtained from unstable specimens, B stable negative curves and C stable-positive curves (see text).

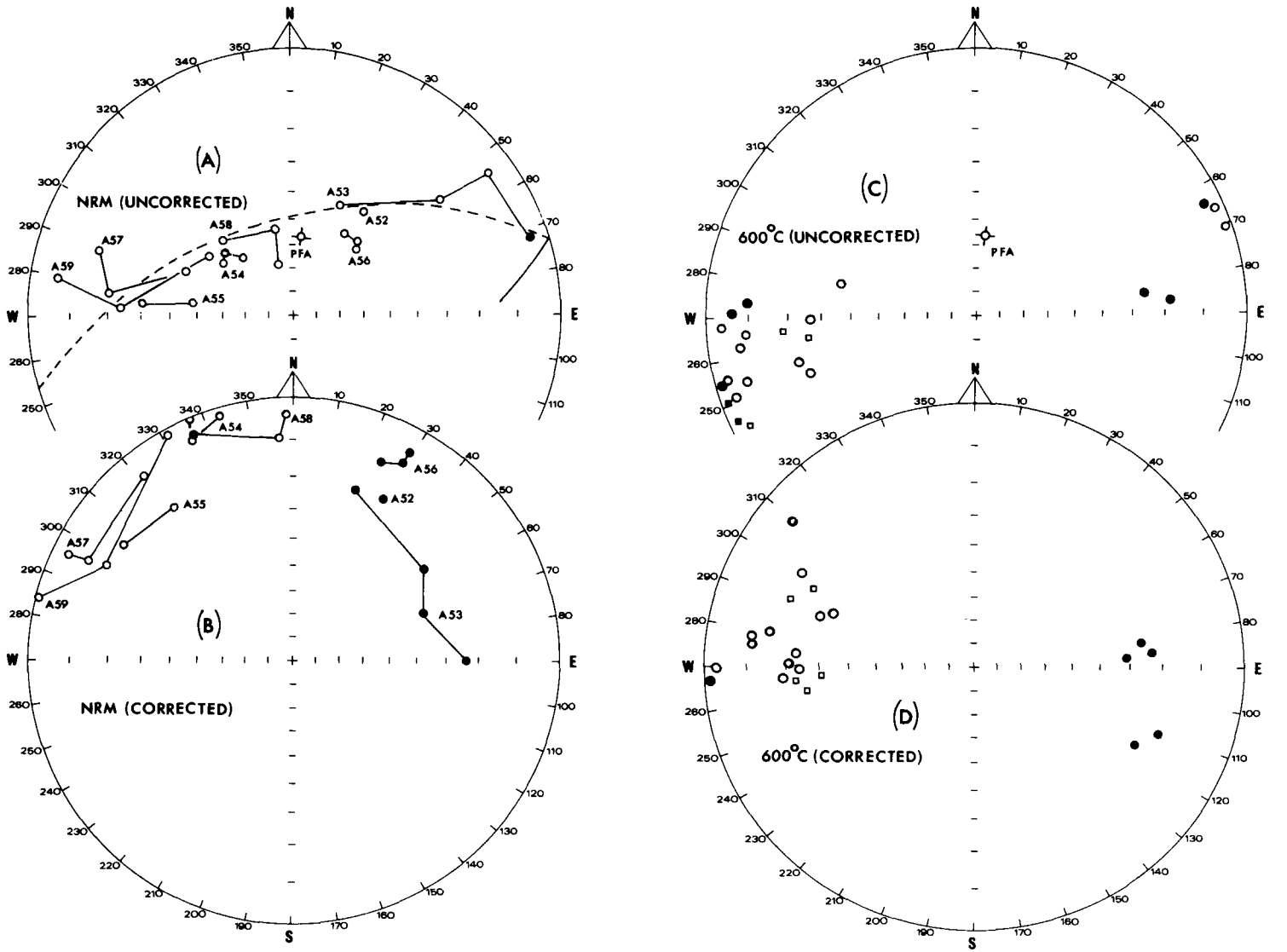


Fig. 5. The Hugh River Shale. Specimen directions before (A and B) and after (C and D) treatment. A and C show the directions referred to the present horizontal (uncorrected) and B and D show them restored to the bedding planes (corrected). Partial thermal demagnetization eliminated initial streaking through the present field axis (PFA). Both normal and reversed polarities were revealed. Square dots in C and D indicate reversed polarization of normally magnetized specimens.

Table 2
Hugh River Shale: Summary of sample-mean data before and after partial thermal demagnetization. Directions are referred to bedding planes.

Sample	NRM Data N	R	D	I	Int. $\times 10^{-6}$ Gauss	Cleaned Data N	R	D	I
A52	1	–	29.5	+ 19.5	4.2	1	–	85.0	+ 23.0
A53	4	3.671	58.5	+ 29.5	3.7	4	3.874	98.5	+ 25.5
A54	3	2.993	339.5	– 2.5	5.5	2	1.992	275.5	– 18.0
A55	2	1.973	314.0	– 16.5	6.2	2	1.919	292.5	– 10.0
A56	3	2.995	27.5	+ 9.0	4.1		unstable		
A57	3	2.916	304.5	– 4.0	7.2	3	2.967	271.0	– 11.5
A58	3	2.940	351.0	+ 3.5	4.6	3	2.977	293.0	– 26.0
A59	3	2.796	303.0	– 6.0	4.9	3	2.931	270.0	– 16.0

Mean of sample-means *: N	R	D	I	α_{95}	Mean of VGP's N	R	lat.	long.	α_{95}
7	6.862	278.0	– 19.0	9.0	7	6.881	11.0°N	37.0°E	8.5

* Directions of samples A52 and A53 were reversed to obtain the mean of sample-means. Symbol notation as for table 1.

fied under stable-negative, behaved in an unusual fashion. Whatever the explanation, the case would seem to be capable of operating at random levels throughout the succession. As described in sect. 2, cross-bedding is extensively developed through the whole sequence. Small-scale features of this nature are readily identified in the field but cross-bedding on a larger scale –say, of the order of several feet in depth–may not be noticed in the small river-bed out-crops, particularly when the bedding is massive. Some of these samples may, then, have been collected from some such horizons. Their inclinations are generally shallow with respect to the true bedding planes (fig. 2B), which might be expected from deposition in a cross-bedded sedimentary environment, if their remanent magnetism was detrital in origin.

The 16 samples which produced consistent results (steep positive inclinations) are assumed to have retained a proportion of their primary remanence: the distribution of their mean directions, after thermal cleaning, is improved, although within-sample scatter often remains high. Their mean direction of magnetization is 027.5° + 78.5° (N = 16), and yields a palaeomagnetic south pole position at 8.5°N 35°W, α_{95} = 25°.

(b) The Hugh River Shale: Samples A52 and A53 (normal polarity) were collected from a lower stratigraphic horizon than samples A54 and A55 (reversed polarity), a 30-ft. thick carbonate band separated the two beds. Samples A56 to A59 (reversed polarity) were collected from a third and higher bed (separated from the A54 and A55 horizon by a 3-ft. thick carbonate band). It is probable that the change in polarity represents a geomagnetic field reversal within the Cambrian period, a result not at variance with other Cambrian polarity data [11]. Stability of the remanent magnetization has been demonstrated in the laboratory by thermal treatment and the presence of a reversal is good evidence for assuming the isolated component to be primary. It is, of cource, appreciated that this result is obtained from only 7 samples; more are required and will indeed be sought; however, it remains uncertain whether the time interval sampled is sufficiently long so that secular variation of the geomagnetic field has been averaged. The result, even if interpreted as a virtual geomagnetic pole (VGP) can be used as a guide to the direction of the geomagnetic field at the time of rock formation. The mean direction for the Hugh River Shale is then, 278.0 – 19.0 (N = 7) and yields a south pole position at 11°N 37°E, α_{95} = 8.5°

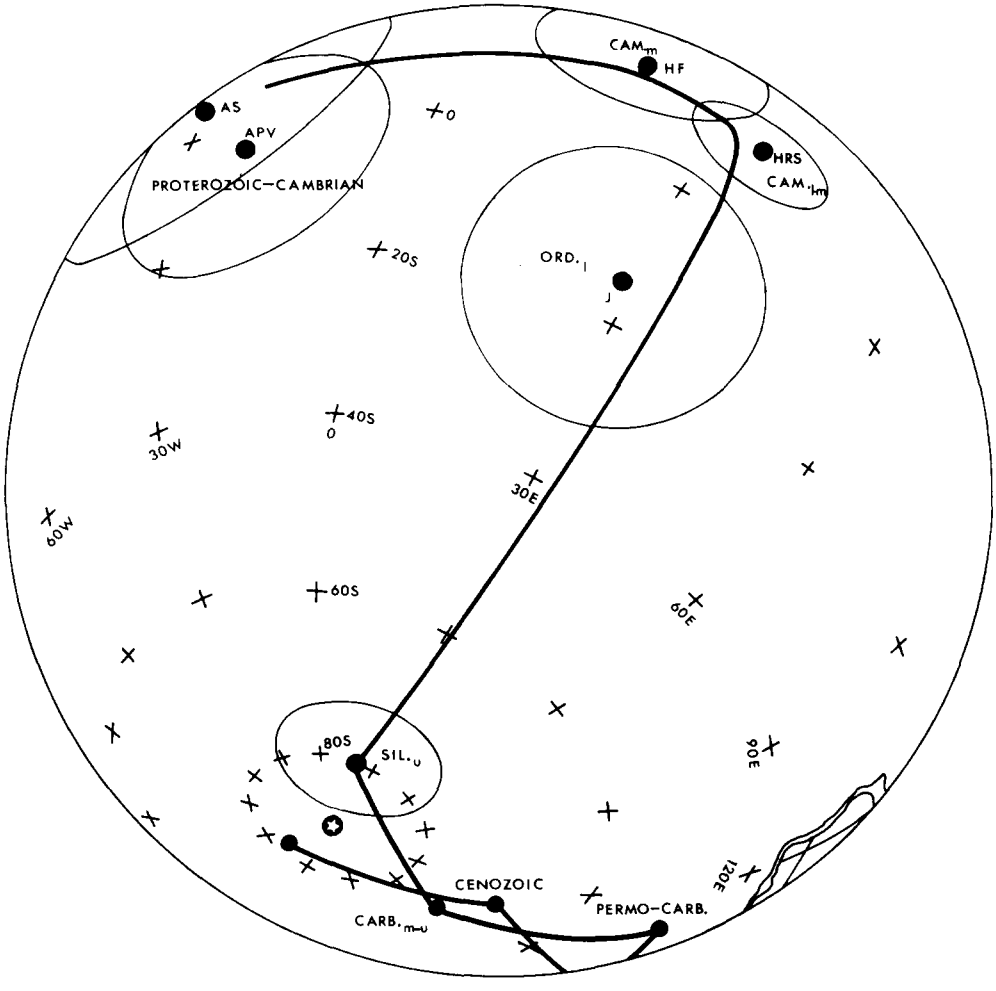


Fig. 6. The polar wander curve for Australia for Phanerozoic time based on data at present available. The west to east Cambrian shift is recorded in the igneous and sedimentary sequence of northern Australia [6, 7] and also in the sedimentary sequence of central Australia (this paper). APV = Antrim Plateau Volcanics, HF = Hudson Formation, J = Jinduckin, AS = Arumbera Sandstone, HRS = Hugh River Shale. The remainder of the curve is based on information contained in ref. [6].

6. Discussion and conclusions

Results from the two rock units indicate that during the Lower–Middle Cambrian an episode of polar wander (and/or plate movement) occurred relative to the continental block which contained the Amadeus Basin. This conclusion is consistent with earlier interpretations [6, 7] based on palaeomagnetic results from northern Australia. The Antrim Plateau Volcanics yield a pole position at 9°S 20°W (α_{95} = 17°) and their age is considered to lie between 663 ± 43 my and the late Lower Cambrian [12]. The Arumbera Sandstone, with an assigned age of late Proterozoic to Lower Cambrian (see sect. 2) is then considered contemporaneous and its pole lies close to the pole of the Antrim Plateau Volcanics. The second pole position recently reported from northern Australia is from a study of the Jinduckin Formation [7]. Its age is considered to be uppermost Cambrian to Lower Ordovician [13] and yields a pole situated at 16°S

29°E (α_{95} = 17°). The Cambrian episode of polar wander therefore appears to have been recorded in the stratigraphic sequences (of sedimentary and igneous type) of northern and central Australia and the Amadeus Basin data suggested that the shift took place rapidly, during the Lower–Middle Cambrian.

One other result from northern Australia which supports this conclusion is that derived from a palaeomagnetic study of the Hudson Formation [14], formerly the Elder Mountain Sandstone [3]. It is the youngest member of the Negri Group of Middle Cambrian sediments and yields a pole position at 18°N 19°E (α_{95} = 13°). Although Luck was uncertain whether the primary component of remanence had been effectively isolated, the pole position is in broad agreement with the foregoing results and would also indicate that the Cambrian polar shift occurred over a short geological interval. The five pole positions are shown in fig. 6. The episode of polar wander (and/or plate movement) amounted to between 50° and 60° of latitude relative to northern and central Australia. It would also seem that these two crustal blocks have maintained their relative geographic positions since at least Early Cambrian times.

Acknowledgements

I am grateful to Dr. M.W. McElhinny for helpful discussion during the earlier stages of the work and to Mr. D.J. Edwards who assisted with sample collection. Both are members of the Australian National University. My thanks also to Dr. A.T. Wells of the Commonwealth Bureau of Mineral Resources, Canberra, for valuable discussion concerning the geology of the sampling locality.

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